ECSE 331
Laboratory No. 1
Introduction to PSPICE

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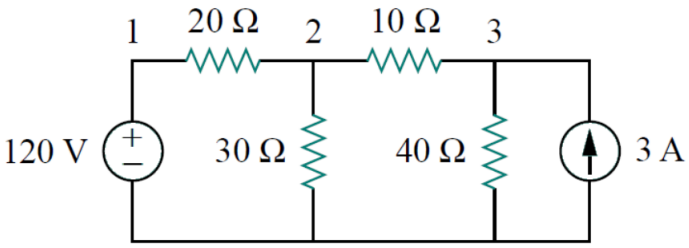
***Abstract*—SPICE (Simulation Program for Integrated Circuits Emphasis) is a general-purpose circuit simulator. In this laboratory, we practice the basics of creating circuit schematics and simulating them in PSPICE using three experiments. The results obtained using PSPICE simulations are then compared with theoretical results obtained by manual circuit analysis. The three experiments consist of finding the voltage at given nodes, the current in given branches, and the Thevenin and Norton equivalent in three circuits. Despite finding the same results using manual circuit analysis, analysis using PSPICE simulation is significantly faster and removes the possibility of calculation errors. Moreover, PSPICE enables us to solve complicated circuits, which would be a very long task if done manually.**

I. Introduction

Experiment 1

This experiment required using PSPICE to find the voltage values at the three nodes (*1, 2,* and *3*) in the given circuit. The given circuit is shown in figure 1.

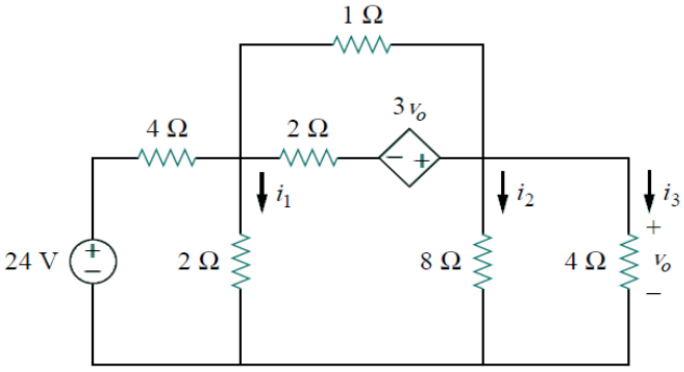
Fig. 1. Exercise circuit



Experiment 2

This experiment required using PSPICE to find the current values in three branches (i_1, i_2, and i_3). The given circuit is shown in figure 2.

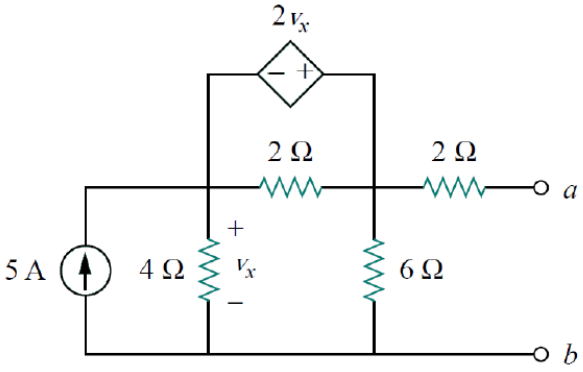
Fig. 2. Exercise circuit



Experiment 3

This experiment required using PSPICE to find the Thevenin/Norton equivalent of a given circuit. The given circuit is shown in figure 3.

Fig. 3. Exercise circuit

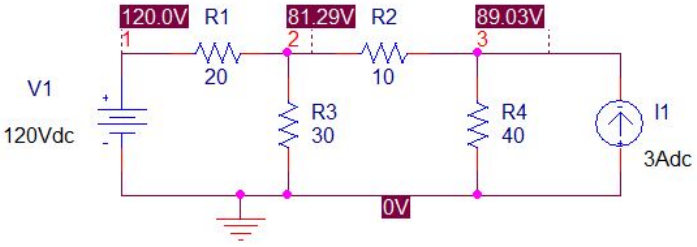


II. Experiments

Experiment 1

We first created a schematic of the circuit given in figure 1. Drawing the schematic of the circuit required the use of four resistors R from the *Analog* library, a *DC* independent voltage source, a *DC* independent current source from the *Source* library, wires, and a ground as can be seen in figure 4. Once the circuit schematic was completed, we performed a *DC* analysis using *DC* bias point simulation in PSPICE to find the voltages at the given nodes. DC bias point voltage simulation results can be seen in figure 4.

Fig. 4. Exercise 1 circuit schematic and simulation results



As figure 4 indicates, the voltage values at nodes *1*, *2* , and *3* were found to be ***120V***, ***81.29V*** ,and ***89.03V*** respectively. Results found using PSPICE DC bias point voltage analysis were then verified by manual circuit analysis. The voltage values at the three nodes found by manual circuit analysis shown in Appendix A are consistent with values found from the PSPICE simulation.

Exercise 2

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Exercise 3

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III. Conclusion

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APPENDIX

Exercise 1

Exercise 2

Exercise 3